# Ajay Venkatesh

Brooklyn, NY

+1 516 585 9013


## Education

**New York University**, New York, NY — *Aug 2024 – May 2026*

*Master of Science in Computer Engineering* (GPA: 3.9/4.0)

*Coursework:* System Design, Machine Learning, Deep Learning, Big Data, Advanced Java, Data Structures

## Experience

- **Amazon** — *May 2025 – Aug 2025*
  *Software Development Engineer Intern* — Seattle, US
  - Engineered the backend services in **Java** (Coral, Smithy, Dagger, CBOR serialization); applied **low-level design patterns**, cutting backend execution time by **35%** with **unit and integration testing** via **Mockito**.
  - Built a high-throughput, event-driven system on **AWS ECS Fargate, SQS, and SNS**, decoupling distributed message processing and reducing response latency by **50%** under peak load.
  - Built a **React 18** frontend with optimized API integration, advanced filtering, and client-side caching, reducing page load time by **40%**.
  - Provisioned **serverless Infrastructure-as-Code** using **AWS CDK** (API Gateway, Lambda, DynamoDB, S3, IAM, CloudWatch), reducing provisioning time by **70%** with **multi-stage, multi-region CI/CD**.
- **Campus Mesh** — *Jan 2026 – Present*
  *Software Engineer* — New York, US
  - Developed scalable **Node.js microservices** powering real-time academic workflows for **10K+ users** across study groups, messaging, notifications, and profile management, backed by a **MongoDB** store.
  - Built a personalized **recommendation engine** using collaborative filtering and behavioral analytics, increasing engagement by **28%** and retention by **50%**.
  - Worked on **LLM-based retrieval pipelines (RAG)** for CampusIQ (AI Assistant), leveraging **embedding-based semantic search** and structured data indexing to improve contextual accuracy by **30%**.
- **New York University (Novel AI Technologies, Inc.)** — *Aug 2024 – Present*
  *Software Developer* — New York, US
  - Architected an **NSF-funded ($100K)** AI-powered health app for **iOS** and **Android** processing real-time wearable data; deployed cloud-hosted AI models for health metric analysis and food image processing.
  - Developed a full-stack insurance analytics platform (**React, Node.js, REST APIs**) deployed via **Docker** on **EC2**; implemented Stripe billing, RBAC, and row-level security, generating **$50K+** revenue across **6 clients**.
  - Engineered **ETL pipelines** transforming **NoSQL** into **PostgreSQL** with **concurrency control and locking**; processed **250K+** records and powered **AWS QuickSight** dashboards.
  - Streamlined underwriting via a broker portal integrated with insurer platforms; implemented **document/media processing** and **WebSocket-based real-time communication**.
- **PricewaterhouseCoopers (PwC)** — *Aug 2021 – Aug 2024*
  *Senior Software Engineer* — Bengaluru, India
  - Developed a **distributed backend** on **Microsoft Azure** for a health platform integrating with **Microsoft CRM**, supporting event-driven processing, fault-tolerant retries, and **privacy-aware handling** of sensitive patient data over **SQL Server** stores.
  - Deployed an **NLP-based OCR neural model** on Azure to automate structured data extraction, reducing processing time by **75%**.
  - Optimized data ingestion with **parallel processing** on **100K-row** feeds, cutting execution from **5 hours to 1 hour**.
  - Led **code reviews** and mentored engineers on **software design principles**, reducing production bugs by **25%**.

## Projects

- **Advanced Expense Tracker (Java Application)** | *Java, JavaFX, JDBC, SQLite, JFreeChart* [GitHub]
  - Architected a modular **Java application** for individual and group expenses with secure login, applying **OOP principles** and **MVC & DAO design patterns**.
  - Implemented concurrent transaction handling and real-time cost splitting using **multithreading** and synchronized blocks for thread-safe DB updates.
- **Flight Delay Prediction (Big Data)** | *Hadoop HDFS, PySpark, Spark SQL, Spark MLlib, Seaborn, Plotly* [GitHub]
  - Built a distributed big data pipeline over **Hadoop HDFS** on **2024 BTS US flight data**; used **PySpark** and **Spark SQL** for cleaning, feature engineering, and large-scale aggregations.
  - Trained **Spark MLlib** classification models for delay severity prediction with feature-importance analysis via Seaborn, Matplotlib, and Plotly.
- **Hospital Management System (Full-Stack)** | *React, Node.js, Express, MongoDB, SageMaker, Hugging Face*
  - Built a **full-stack healthcare platform** (React, Node.js, Express, MongoDB) managing appointments, patient records, and lab reports with **role-based access control**.
  - Fine-tuned a **Hugging Face transformer-based model** reaching **88% accuracy** for health condition prediction and deployed on **AWS SageMaker**.

## Skills

- **Programming Languages:** Java, Python, C++, TypeScript, JavaScript (ES6), SQL, Bash
- **Backend & Distributed:** Microservices, REST API, GraphQL API, Apache Spark, PySpark, Spark SQL, Hadoop, ETL Pipelines, Event-Driven Systems
- **Web & Frontend:** React, Node.js, Express.js, HTML, CSS, Sequelize
- **Databases:** PostgreSQL, MySQL, MongoDB, DynamoDB, SQL Server, Redis
- **Cloud & DevOps:** AWS (CDK, Lambda, ECS Fargate, SQS, SNS, S3, DynamoDB, API Gateway, CloudWatch, IAM, SageMaker), Azure (Function Apps, Storage), Docker, Kubernetes, CI/CD
- **AI/ML:** LLM Fine-tuning, RAG (LangChain, FAISS), Transformers, NLP, Hugging Face, PyTorch, TensorFlow
- **Tools & Platforms:** Git, Linux, Visual Studio, Cursor, GitHub Copilot, Jupyter, Postman, Mockito

## Publications

Supervised ML System Based Segmentation and Classification of Stroke Using Deep Learning Techniques *(IEEE)*

An Aquila-optimized SVM Classifier for Diabetes Prediction *(IEEE)*

## Certifications

AWS Cloud Practitioner | Stanford Machine Learning (Coursera) | Azure Data Fundamentals